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Simulation-Based Inference for the Star-Forming Interstellar Medium: Inferring Physical Parameters from PHANGS Observations

Chang-Goo Kim[1]

[1] *Department of Astrophysical Sciences, Princeton University, 4 Ivy Lane, Princeton, NJ 08544, USA*

ABSTRACT

PLACEHOLDER: Abstract to be written once the inference framework is implemented. Key points: ISM parameter inference problem, why explicit likelihoods are unavailable, SBI/ILI as solution, connection to Paper I simulation suite, expected inference targets and posterior diagnostics.

Keywords: Interstellar medium (847); Star formation (1569); Stellar feedback (1602); Magnetohydrodynamical simulations (1966); Methods: statistical (1361)

1. INTRODUCTION

Resolved multi-wavelength surveys of nearby star-forming galaxies, most prominently the Physics at High Angular resolution in Nearby GalaxieS (PHANGS) survey (A. K. Leroy et al. 2021; D. R. Weisz et al. 2023), deliver aperture-level joint constraints on the cold gas, stellar, and star-forming components of the interstellar medium (ISM) across ~ 80 galaxies and more than two orders of magnitude in environment. These observations have established empirical scaling relations between the local dynamical equilibrium pressure $P_{\rm DE}$, the molecular gas fraction $f_{\rm mol}$, and the star formation rate surface density $\Sigma_{\rm SFR}$ (J. Sun et al. 2020, 2023; V. Vijayakumar et al. 2025), which are reproduced in physical terms by the pressure-regulated, feedback-modulated (PRFM) framework (E. C. Ostriker & C.-G. Kim 2022). PRFM replaces a single empirical Kennicutt–Schmidt-type prescription with a small set of physical parameters—feedback yields, effective velocity dispersion, phase fractions—that connect resolved ISM observables to the microphysics of feedback. The central inference problem this paper addresses is the inverse one: given a vector of PHANGS aperture observables, what is the posterior distribution over the physical parameters that govern ISM behavior, jointly with the environmental parameters that describe the local galactic context?

This inverse problem is hard for a reason that is structural rather than incidental. The forward map from environmental and physical parameters to observed aperture summaries is mediated by a fully nonlinear, multiphase, turbulent system whose snapshot realizations are stochastic. Even at exact input parameters, the observable output of an TIGRESS-NCR-class simulation is a single draw from a distribution whose moments—let alone whose full likelihood—are not analytically accessible. The likelihood $p(\boldsymbol{x}_{\rm obs} \mid \boldsymbol{\theta})$ required for Bayesian inference therefore has no closed form, and likelihood-based MCMC is not available. This is the canonical setting of *simulation-based inference* (SBI), also termed likelihood-free inference, in which the unknown likelihood is replaced by an implicit one defined by a stochastic simulator (K. Cranmer et al. 2020).

Modern SBI trains a neural density estimator to approximate either the posterior $p(\boldsymbol{\theta} \mid \boldsymbol{x}_{\rm obs})$ directly (neural posterior estimation), the likelihood $p(\boldsymbol{x}_{\rm obs} \mid \boldsymbol{\theta})$ (neural likelihood estimation), or the likelihood ratio between competing parameter values (neural ratio estimation), using $(\boldsymbol{\theta}, \boldsymbol{x}_{\rm sim})$ pairs drawn from the simulator (K. Cranmer et al. 2020). Once trained, the estimator amortizes the inference: posteriors can be evaluated at arbitrary observed data without re-running the simulator. SBI has been applied at scale to cosmological parameter inference, galaxy-formation physics, and gravitational-wave parameter estimation CGK: (citations to add: Alsing+19, Cranmer+20, Dax+21, Hahn+22 SimBIG, Villaescusa-Navarro+23 CAMELS-SBI; verify ADS keys before inserting). In each case, the strategy succeeds when a structured simulation ensemble covers the parameter volume densely enough to train the estimator, and when the summary statistics chosen preserve the information that distinguishes parameter values. Applications of SBI to the resolved, multiphase, star-forming ISM have so far been limited; this paper, to our knowledge, is the first to deploy SBI on PHANGS aperture observables using a structured TIGRESS-NCR simulation suite as the simulator. CGK: Verify the novelty claim; soften to "among the first" if a closely comparable effort exists.

The simulation suite required for this program is the subject of the companion paper CGK: (?; Paper I, hereafter ?). ? constructs a PHANGS-informed simulation design for the TIGRESS-NCR-NCR framework (C.-G. Kim et al. 2024): a kernel density estimator is fit to the joint PHANGS aperture distribution of five environmental parameters—the gas surface density $\Sigma_{\rm gas}$, stellar surface density $\Sigma_{\rm star}$, stellar scale height $H_\star$, orbital frequency Ω, and shear parameter q—and a scrambled Sobol quasi-random sequence is mapped through its marginal quantile functions to generate the design. From each simulation we extract summary statistics that correspond directly to PHANGS aperture observables: $\Sigma_{\rm SFR}$, $f_{\rm mol}$, $P_{\rm DE}$, the effective vertical velocity dispersion $\sigma_{\rm eff}$, and the thermal and non-thermal feedback yields $\Upsilon_{\rm th}$ and $\Upsilon_{\rm turb+mag}$. The target of inference in this first paper is the posterior $p(\boldsymbol{\theta}_{\rm env} \mid \boldsymbol{x}_{\rm obs})$ over the five-dimensional environmental parameter vector, conditioned on fiducial physics ($Z'_g = 1$, $Z'_d = 1$, standard IMF, calibrated TIGRESS-NCR-NCR feedback). The framework is designed so that the physics parameter vector $\boldsymbol{\theta}_{\rm phys}$ can be appended once the physics-extension block described in ? is run.

The scope of this paper is the construction, validation, and first application of the inference framework. We present (i) the choice of summary statistics and the architecture of the neural posterior estimator; (ii) training on the environmental simulation suite of ?; (iii) posterior recovery and coverage diagnostics on held-out simulations and on mock observations drawn from the simulator; and (iv) application to PHANGS aperture measurements to infer $\boldsymbol{\theta}_{\rm env}$ posteriors for each aperture and to test the consistency of inferred parameters across galaxies and across radial bins. Two extensions are deliberately deferred to future work. The first is joint inference over $\boldsymbol{\theta}_{\rm env}$ and $\boldsymbol{\theta}_{\rm phys}$, which requires the physics-variation block of the suite. The second is field-level inference using resolved PHANGS maps, which requires the synthetic observation pipeline outlined in ? together with non-Gaussian summary statistics such as wavelet scattering coefficients (?). Both extensions are designed to slot into the present framework without architectural changes. CGK: Confirm this is the intended scope for Paper II; in particular, whether posterior recovery on mock data and the PHANGS application are both in scope for this paper or whether the PHANGS application is itself deferred.

The remainder of this paper is organized as follows. CGK: Section 2 describes the PHANGS aperture observables used as inference targets and the joining procedure between the megatable and matched ancillary catalogs. CGK: Section 3 summarizes the TIGRESS-NCR-NCR simulation suite of ? as it is used here—the implicit prior $p(\boldsymbol{\theta}_{\rm env})$, the simulator $p(\boldsymbol{x}_{\rm sim} \mid \boldsymbol{\theta}_{\rm env})$, and the choice of summary statistics. CGK: Section 4 presents the inference framework: the neural posterior estimator, the training procedure, and the coverage and calibration diagnostics used to validate it. CGK: Section 5 presents the inferred posteriors on held-out simulations and on PHANGS apertures, and discusses the implications for ISM physical parameters. We conclude in CGK: Section 6. CGK: Section labels assumed from the existing sections/ directory listing (data, simulation_prior, inference, results, conclusions); adjust if the final structure differs.

2. OBSERVATIONAL DATA

The observational constraints for the inference framework are drawn from the same PHANGS megatable dataset described in Paper I (Kim et al., in prep.; J. Sun et al. 2022, 2023). We refer the reader to Paper I for details of the data reduction, aperture definitions, and derived quantities.

For the inference problem, the relevant observables at each aperture are:

$$\mathbf{x}_{\rm obs} = (\Sigma_{\rm gas},\, \Sigma_{\rm SFR},\, f_{\rm mol},\, P_{\rm DE},\, \sigma_{\rm eff}) \tag{1}$$

where $\Sigma_{\rm SFR}$ is the star formation rate surface density, $f_{\rm mol} = \Sigma_{\rm mol}/\Sigma_{\rm gas}$ is the molecular fraction, $P_{\rm DE}$ is the dynamical equilibrium pressure, and $\sigma_{\rm eff}$ is the effective ISM velocity dispersion. [SKELETON — summary statistics and completeness masks to be documented here]

3. SIMULATION SUITE AS PRIOR AND SIMULATOR

[SKELETON] The TIGRESS-NCR-NCR simulation suite from Paper I provides both the prior distribution over environmental parameters and the numerical simulator that maps parameters to observable outputs.

3.1. *Simulation Inputs and Sampling*

The environmental prior is the PHANGS-informed KDE-Sobol design described in Paper I. The suite spans $\boldsymbol{\theta}_{\rm env} = (\Sigma_{\rm gas}, \Sigma_{\rm star}, H_\star, \Omega, q)$ at fiducial metallicity $Z'_g = 1$.

3.2. *Observable Summary Statistics*

For each simulation, we extract the following summary statistics that can be compared to PHANGS aperture measurements:

- $\Sigma_{\rm SFR}$: star formation rate surface density
- $f_{\rm mol}$: molecular fraction
- $P_{\rm DE}$: dynamical equilibrium pressure
- $\sigma_{\rm eff}$: effective ISM velocity dispersion
- Feedback yields $\Upsilon_{\rm tot}$, $\Upsilon_{\rm th}$, $\Upsilon_{\rm turb+mag}$

[SKELETON — summary statistic extraction pipeline to be documented]

3.3. *Mapping to PHANGS Observables*

[SKELETON — aperture convolution and observational noise model]

4. INFERENCE FRAMEWORK

4.1. *Inference Targets*

[PLACEHOLDER] The inference problem is: given observed PHANGS summary statistics $\mathbf{x}_{\rm obs}$ at an aperture, infer the posterior distribution over the environmental and physical parameters $\boldsymbol{\theta}_{\rm env}$ (and eventually $\boldsymbol{\theta}_{\rm phys}$).

4.2. *Neural Posterior Estimation*

[PLACEHOLDER] We use a neural posterior estimator (NPE) trained on simulation–observation pairs $(\boldsymbol{\theta}_{\rm env}^{(k)}, \mathbf{x}^{(k)})$ from the TIGRESS-NCR-NCR suite.

[Architecture, training procedure, and hyperparameters TBD]

4.3. *Training and Validation*

[PLACEHOLDER]

4.4. *Posterior Diagnostics*

[PLACEHOLDER] Simulation-based calibration (SBC), coverage tests, expected coverage probability (ECP) checks.

5. RESULTS

5.1. *Posterior Recovery on Mock Data*

[PLACEHOLDER] We validate the inference framework on mock data generated from held-out simulations. Summary calibration diagnostics and parameter recovery tests are reported here.

5.2. *Application to PHANGS*

[PLACEHOLDER] Posterior distributions over $\boldsymbol{\theta}_{\rm env}$ conditioned on observed PHANGS aperture statistics.

5.3. *Degeneracies and Identifiability*

[PLACEHOLDER] Posterior correlations between environmental parameters; identifiability of physics parameters once the physics suite is available.

6. CONCLUSIONS

[SKELETON] We have presented a simulation-based inference framework for the star-forming ISM, using the TIGRESS-NCR-NCR simulation suite from Paper I as the numerical prior and simulator, and the PHANGS megatable as the observational constraint.

Key results:

PLACEHOLDER

PLACEHOLDER

Future extensions include application to the physics parameter suite (metallicity, IMF, feedback variations), field-level inference using wavelet scattering statistics, and comparison with synthetic observations from a forward-modeled multiwavelength pipeline.

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